



CSC1112: Domain Applications 3

Module Co-ordinator: Sahraoui Dhelim

Scientific Paper Assignment

Note: This assignment is worth 80% of your overall grade for the module.

Submission

1. Paper
2. Presentation

Deadline

5th April 2026 @ 23:59
Last week of Sem2

Groups: This assignment can be done in 1-4 students team

0 Introduction

In this assignment, you will be required to use the insights that you have gained from the presentations in this module in order to write a scientific paper that propose a novel solution or/and review existing literature, exploring the application of **Data Science and Artificial Intelligence (AI)** in a given application domains, e.g. **Agriculture, Healthcare, Sports...etc.**

This assignment has three sections:

1. **Paper:** In this document, you will outline a real-world problem you want to solve or an overarching question you would like to answer with through using AI and data analytics technologies within a given application domains. You will also be required to outline the relevant tasks/research questions, choose a dataset or set of datasets to address the tasks outlined, assess the legal details of using your chosen data and consider the ethical implications of your chosen tasks.
2. **Presentation:** Here, after finishing your project, you will present it.

This assignment has a total of **80 marks**. **Every calendar day late will incur a 5% penalty**, except in the case of extenuating circumstances as outlined in the university policy.

1 Paper (75 marks)

A template for the paper is provided below along with a marking rubric which sets out a clear set of guidelines for the expected content of the report. You should adhere to this in order to obtain a good mark. The use of Generative AI (e.g. DeepSeek/ ChatGPT) is permitted; however, their use must be clearly declared, with a detailed explanation of how they were used. Failure to disclose the use or misuse of Generative AI will result in the relevant sections being considered void. Part of this project is to develop your own writing and communication skills, so please take this opportunity to develop these skills independently.

1.1 Paper type

The paper can take one of the following approaches:

1. **Proposal of a Novel Solution** – Introduce a new idea or framework using AI and Data Science to solve a problem in the chosen domain.
2. **Literature Review** – Analyse and summarize existing research on a specific application of an AI technology in the chosen domain.

1.2 Research Aims

Outline a theatrical or practical problem that you want to solve/discuss. The **Research Aims** section outlines the **main objectives** of the study. It defines **what the research intends to achieve** and provides a clear direction for the paper. This should take the form of a problem title and **research questions** that you would like to answer or tasks you would like to complete.

1.3 Dataset

If your paper involves an experimental section, choose data from a **reputable source**, or one the datasets provided by the module coordinator

2 Presentation (5 marks)

Regardless of the presentation format, you should include the following information.

- Problem Description
- The solutions to this problem that are out there already, if any.
- Brief description of dataset
- Proposed Approach
- Result of Analysis

In-class presentation

The presentation should be roughly 10 minutes long. There will be an additional 3-5 minutes for questions afterwards.

3 Template

You are provided with an Overleaf project, where you should write your paper. The paper should take the following structure:

The Title of Your Project

YourNames
Student Number: 123456789
School of Computing
Faculty of Engineering and Computing
Dublin City University
youremail@dcu.ie

Abstract

Describe your project and conclusions from a high level (max 150 words, 5 marks).

1 Introduction

(estimated 750 words, 15 marks)

2 Related Work

(estimated 750 words 15 marks)

3 Methodology

(estimated 500 words, 30 marks)

4 Experiments

(estimated 500 words, 20 marks)

5 Results

(estimated 500 words, 15 marks)

6 Conclusion & Future Work

(max 300 words, 5 marks)

7 Ethics statement

8 Reference

